An Alternative To Deduction

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Abstract

Deductive Representations are well defined, easily inspected, and precise. However, they are also brittle, inflexible and difficult do debug. We propose an alternative *plausible* representation which which is weaker than deduction. It can accept and reason with knowledge which is less precise, and there is no notion of global consistency, only local consistency of an explanation. Like connectionism and unlike deductive approaches this approach gains strength from empirical data.

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1. PI–EBL APPROACH

There are many advantages to a reasoning model based on a declarative representation of world knowledge. Such a representation may be leveraged by many knowledge intensive mechanisms: it can be communicated to other agents, multiple models can be combined, parts of a model can be isolated and independently verified, etc. Because deduction is well understood it is natural to ground these models in a deductive framework. Requiring all knowledge to be expressed in a deductive form is restrictive, and causes systems to be very brittle. Connectionist approaches, on the other hand, are not based on such a restricted language, which is a source of their flexibility [Rumelhart86]. The symbolic/connectionist distinction may appear unimportant since the sources of knowledge potentially available are the same. However, research in connectionism is impeded by its representation—its very difficult to understand what abstract principles are revealed by a connectionist solution to a problem. The advantages of both approaches can be achieved by a reasoning paradigm based on an explicit world model (like symbolic approaches) using a non–deductive representation (like connectionist approaches). Research is facilitated by such a paradigm, high–level knowledge intensive mechanisms can be more easily constructed, and the principles underlying their success are much more available for inspection. In this paper we introduce a logic of plausible reasoning with these desired properties. As in distributed connectionist systems, our approach crucially involves learning through observations of the world. The machine learning paradigm used to support our logic of plausible reasoning is EBL.

Explanation Base Learning is a learning mechanism which combines a model of the world with examples from the world to construct explanations which can be later used to reason about the world [DeJong86, Mitchell86]. Like other knowledge intensive mechanisms, it gains its strength from its explicit model. Much of EBL's potential as a methodology has not been tapped, however, because the knowledge representation it uses is too restrictive. For example, EBL's restriction to symbol level learning is due to the base representation, *not* an inherent restriction on the EBL paradigm. (Symbol level learning adds no new knowledge but simply makes some knowledge much easier to derive [Dietterich86].) EBL's limited use of its training examples is also traceable to its base representation. In general the flexibility of any architecture depends, in large part, on the representation language it uses, no matter what high level task is being performed. We present a plausible representation and inferencing mechanism as an alternative to classical deduction. The flexibility of this representation is demonstrated by using the Plausible Inferencer as the explanative component for an Explanation Based Learner (PI–EBL). Through its flexible representation PI–EBL overcomes EBL's deficiencies mentioned above, and acquires some advantages associated with both connectionism and traditional induction by greater use of its training examples.

2. MOTIVATION OF PLAUSIBLE DOMAIN THEORIES

There are three important properties inherent in any model of a domain. We will use these properties to demonstrate the inadequacies of deductive knowledge representations. These properties also are used as a guide in the selection of an appropriate plausible representation. The first property, *simplicity*, is a measure of the amount of inference needed to derive a conclusion about the domain from the domain theory. This is often related to the syntactic simplicity of the domain theory. At times we will use this as a secondary definition of simplicity. The second property, *coverage*, is a measure of the number of conclusions about the domain which can be drawn from the domain theory. So a theory with low coverage makes no predictions about large parts of the domain. The third property, *faithfulness*, is a measure of the chances that a statement derivable from the domain theory is consistent with the domain itself. An unfaithful domain theory makes incorrect predictions about its domain. In understanding these terms it is helpful to notice the analogy: faithfulness to soundness and coverage to completeness. We avoid these more familiar terms in our discussion because their meaning within model theory [Genesereth87] is more constrained than the meaning intended in this paper. In particular we do not wish to constrain our domain theories to be theories which are modeled by their domain in the model theoretic sense. In fact, we are anticipating domain theories which are not even satisfiable (internally consistent); obviously if there is *no* model of our domain theory the domain itself could not be such a model.

Simplicity, coverage, and faithfulness are all terms which describe how well a theory captures a domain. An optimal theory is one that maximizes all three aspects of the theory. Often this turns out to be a tradeoff, theories which maximizes one property do so at the expense of another. This is best demonstrated through example. Consider theories that describe things which are able to fly. One possibility is: $Bird(x) \rightarrow Fly(x)$. This is both very simple, and has pretty good coverage. Another possibility, $Animate(x) \wedge Bird(x) \rightarrow Fly(x)$, trades some of this simplicity for greater faithfulness. It takes more computational effort to derive $Fly(x)$, and the knowledge base is syntactically more complex, but it correctly handles the dead bird case. Adding $HasFeathers(x) \rightarrow Bird(x)$ to the knowledge base will increase coverage since we now infer $Fly(x)$ even in the case that we don't know x is a bird, but only that it has feathers. This increase is at the expense of both simplicity, adding another inference step, and faithfulness, not all feathered things are birds. Faithfulness, simplicity, and coverage form a tradeoff: the optimal choice for this tradeoff is a compromise which does not place too much importance on any one of the three attributes; ignoring any aspect in favor of the others has diminishing returns after a point.

Deductive frameworks are not sufficient as a knowledge representation mechanism. Their deficiency stems from their precision (a non–optimal simplicity/faithfulness tradeoff). Knowledge placed in a deductive framework must be valid—true for all possible examples. This is a problem since much of the knowledge an intelligent agent must deal with is true for most but not *all* examples, and such knowledge cannot be used in a deductive framework. The impossibility of representing real world knowledge in a deductive theory is noted by McCarthy as the qualification problem [Genesereth87, McCarthy69]. The qualification problem states that any universally quantified implication will need a large (infinite) number of preconditions to exclude *all* possible exceptions. The reader may be convinced by considering the implication: $Bird(x) \rightarrow Fly(x)$. Correcting this rule to handle exceptions would require adding $Alive(x)$, $\neg Penguin(x)$, $Sane(x)$, etc. as conjuncts. Of course such a deductive theory is not practical. The traditional approach to this problem is to translate the original domain into a micro–world which is consistent with the original domain to a fixed level of detail. Because the micro–world does not reflect all details of the real world, it is finitely describable using a deductive theory, thus a deductive theory may be employed. The disadvantage to this approach is that the model is really a model of the micro–world *not* a model of the original domain. The agent using such a theory has no recourse if some important portion of the domain is not modeled in the micro–world. Thus, the micro–world must be complex enough to handle the most detailed example encountered by the agent. This forces the entire system to represent and reason using this detailed and complex theory for *all* examples, even those which can be handled using a much simpler theory. Adapting the theory to handle one additional special–case example could easily result in many previously tractable problems to becoming inaccessible. Furthermore, simple theories which make different assumptions (based on different micro–worlds) cannot be directly combined because the resulting theory would likely be internally inconsistent. This is a stifling restriction: different micro–worlds (consistent micro–theories) will be useful in reasoning about different problems, and combining inconsistent micro–theories into a *single* consistent micro–world will result in a theory which is too precise. It is too precise because the closure of these micro–theories contain many facts inconsistent with other micro–theories, and the combined theory must be consistent with only *one* of these choices. Because the combined theory is deductive it must specify *exactly* how its component theories interacted, even when the interactions are uninteresting parts of the deductive closure. Using a micro–world forces faithfulness to be traded for simplicity *even when the faithfulness gained only disambiguates cases which are unimportant in practice*. No deductive chain of inference no matter how complex may be inconsistent with any other over the entire theory. Requiring the theory to be internally consistent without gaining consistency with the world gains us little power at the great expense of scaleability. An agent using a deductive knowledge base cannot accept *any* knowledge about the world until all its interactions with the *closure* of the agents knowledge are precisely specified. This restriction is unacceptable. Avoiding this restriction and its fixed level of detail forces us to abandon the micro–world approach.

Non–monotonic theories provide a mechanism for reasoning with incomplete knowledge [McCarthy86, Reiter80]. McCarthy's *Abnormal*(x) predicates allows one set of rules which derives a conclusion to override another rule. Like deductive theories, however, the sentences entailed by a nonmotonic theory is still precisely defined without any examples being needed to disambiguate inconsistencies. The whole notion of circumscription makes precise the closure of knowledge. In order for a representation to be an effective alternative to the micro–world approach, it cannot be deductive; it must incorporate a means of trading between faithfulness, simplicity, and coverage. This means it cannot allow the closing of the knowledge base. Instead the world must be left open. Inferred knowledge should be used as *suggestions* to be empirically verified, rather than theorems with are necessarily true.

3. REPRESENTATION OF PLAUSIBLE DOMAIN THEORIES

Plausible domain theories are syntactically similar to a set of implications. Semantically they are quite different, however, since the theories are not deductive. The theories must *suggest* possible relationships between concepts without *entailing* them. Semantically the plausible domain theory may be interpreted as deductive implications with missing preconditions. The missing preconditions are collectively called the *implicit context* of the implication. The implicit context is the set of additional constraints sufficient to guarantee that the plausible implication deductively holds. The preconditions of a plausible implication are not sufficient for concluding their consequent. Thus we refer to plausible implications as *influents* to emphasize the idea that the consequent is influenced, but not entailed, by the preconditions. Asserting an influent specifies that its preconditions are relevant to determining its consequent in some contexts*. An influent also specifies the direction of influence between the preconditions and consequent, and they may be positively or negatively related. These influents provide the basis for representing PI–EBL domain theories.

4. PLAUSIBLE INFERENCING

To build explanations from a plausible domain theory (a set of influents) we need an inference mechanism. Influents may be combined with other influents in a way analogous to deductive implications. The standard mechanism of unifying a precondition of one rule with the consequent of another is not sufficient alone however. An additional inference rule allows combining several influents with the same consequent into an *influent set*. Thus, it is possible to conjecture a complete explana-

* The influent determines its consequent as in relevance logics while allowing the axiom set to remain open as intuitionistic logics.

Influent: $P > Q$

Meaning: $I \wedge P \rightarrow Q$

Implicit Context

Figure 1: Semantics of an Influent.

tion which has all preconditions relevant to determining its consequent. *Plausible explanations* are generated by repeated application of these two inference mechanisms. These explanations are analogous to Stuart Russell's determinations [Grosof], since both serve as a specification of the set of all determining factors for some value in the domain. A plausible explanation specifies that its leaf preconditions determine its consequent, without specifying the mapping. Unlike Russell's determinations, however, there is no guarantee that a plausible explanation actually *does* determine its consequent because nothing is known about its implicit contexts. So a plausible explanation must be empirically verified by checking that it is consistent with examples from the domain before it can be accepted as a plausible theorem. Many different explanations can be conjectured for the same observed behavior. Ideally, the most plausible explanations are generated first. If these are empirically rejected (ie. the observed data contradicts them) the next most plausible explanation is hypothesized and so on until some hypothesis is empirically confirmed. Generation of plausible *theorems* is more complex than its deductive counterpart since is combines both analytical and empirical constraints. Thus, machine learning is tightly intertwined with reasoning.

The purpose of the plausible representation and inference mechanism is to provide a basis for reasoning with imprecise information which allows the reasoner to skirt much of the complexities of deductive reasoning. Since we are proposing this as a natural basis for reasoning it is crucial that our understanding of this representation is precise and that we *precisely* define the meaning of an explanation in this formalism. So we formalize the notion that a plausible theorem is a plausible explanation which is confirmed by observed examples. Precisely, a plausible theorem contains influent sets which satisfy the following three properties relating it to the set of observations:

Let O be the set of observations.
Let A and D be the preconditions of the affirming and denying influents in the influent set I .
Let $P \equiv A \cup D$.
Sat(predicate, observation) is true iff the predicate was satisfied in the particular observation.

Def: I is *deterministic* on O iff

$$\forall o_i, o_j \in \mathrm{O} \quad \left(\forall p \in P \quad Sat(p, o_i) = Sat(p, o_j)\right) \rightarrow \left(Sat(c, o_i) = Sat(c, o_j)\right)$$

The consequent must be completely determined by the preconditions of the influent set with respect to the set of observations. There cannot be two observations which agree on the truth value of all of the preconditions but do not agree on the truth value of the consequent. This is the same constraint which Russell's determinations have.

Def: I is *predictive* on O iff

$$\neg\,\exists o_i, o_j \in \mathrm{O}; A' \subseteq A$$
$$\Big(\forall p \in P \backslash A' \quad Sat(p, o_i) = Sat(p, o_j)\Big) \wedge$$
$$\Big(\forall a \in A' \quad \neg\, Sat(a, o_i) \wedge Sat(a, o_j)\Big) \wedge Sat(c, o_i) \wedge \neg\, Sat(c, o_j)$$

and

$$\neg\,\exists o_i, o_j \in \mathrm{O}; D' \subseteq D$$
$$\Big(\forall p \in P \backslash D' \quad Sat(p, o_i) = Sat(p, o_j)\Big) \wedge$$
$$\Big(\forall a \in D' \quad \neg\, Sat(a, o_i) \wedge Sat(a, o_j)\Big) \wedge \neg\, Sat(c, o_i) \wedge Sat(c, o_j)$$

The second consistency property is predictiveness. When an influent is needed to distinguish between two observations, the influence it has on the consequent must be in the predicted direction. If two observations differ only in the truth values of the preconditions of several positive influents then the consequent must be true when those preconditions are true. This constraint ensures that positive values assigned to the preconditions can only have a positive effect on the consequent. The analogous negative relationship must also hold for negative influents.

Def: I is *relevant* on O iff

$$\forall p \in P \quad \exists o_i, o_j \in \mathrm{O}$$
$$\Big(\forall r \in P \backslash p \quad Sat(r, o_i) = Sat(r, o_j)\Big) \wedge$$
$$\big(Sat(p, o_i) \neq Sat(p, o_j)\big) \wedge \big(Sat(c, o_i) \neq Sat(c, o_j)\big)$$

The final consistency property is relevance. An influent set is relevant over a set of observations if the preconditions of each influent are necessary for uniquely determining the value of the consequent. In effect no influent superfluous to the set of observations is allowed; each influent must be the determining factor in some circumstance otherwise it could be eliminated from the explanation with no effect on the learned function. Relevance is required since it disallows plausible theorems which are correct in the sense that their preconditions determine their consequent, but are useless since they have so many irrelevant influents. Any explanation that satisfies these three properties determines a function which can be used to model the domain. This is PI–EBL's ultimate goal. Thus the PI–EBL approach is a search for plausible theorems—explanations which are empirically shown to be deterministic, predictive, and relevant.

5. REASONING WITH PI–EBL

Examples are much more central when learning is based on explanations from plausible theories. The PI–EBL approach has aspects of traditional induction, but it is not a hybrid system in the sense that it can be easily decomposed into these separate components. Because of the homogeneity of the approach, aspects of both approaches are intertwined in its learning. The plausible inferencer is used to explain the value of the predicate of interest in the system's observations. Observations are also used to verify or reject hypotheses generated by the inferencer. This is done in a generate and test fashion. If an observation presented to the system causes failure of the current hypothesis to be deterministic, predictive, or relevant over the set of observed examples, then the hypothesis is rejected and the inferencer is used to generate another plausible explanation. As plausible explanations are being generated and monitored for correctness the observed examples are used for yet another purpose. For every influent set in the plausible explanation there is a function mapping the the preconditions of the influent set to its consequent. Observations are used to constrain these functions. This use of the observations is more similar to traditional induction since they are used to learn a function with fixed attributes (preconditions of the explanation). Even here, however, the explanative structure is relevant since it constrains the function being induced; it must be consistent with the plausible explanation.

The plausible inferencer builds explanations of the observation in much the same way that traditional EBL explains the observations it is given. Unlike EBL, the PI–EBL approach admits multiple incompatible explanations of the same goal concept. Because of their implicit context, the predictiveness of these explanations will vary greatly, which places great importance on the *order* that the plausible inferencer uses in generating its explanations. Since these explanations bias PI–EBL's learning, constraints on their order provide an ordering bias for the entire PI–EBL system. The a priori chance that an explanation is a theorem can be approximated in several ways. The size of the explanation is useful, larger explanations generally must satisfy a larger implicit context so they are more likely to fail. Another promising estimator of predictiveness can be derived by combining the predetermined predictiveness of each of the influents in the explanation.

6. LEARNING EXAMPLE

The PI–EBL learning algorithm is being implemented. We present a simple example of its execution to supplement its description and as a concrete example of its advantages over traditional EBL.

$$Clawed(x) \succ Bird(x)$$
$$Feathered(x) \succ Bird(x)$$
$$Immobile(x) \bar{\succ} Fly(x)$$
$$Dead(x) \succ Immobile(x)$$
$$Bird(x) \succ Fly(x)$$

Figure 2: Plausible Domain Theory

Figure 2 shows a plausible domain theory which is used to predict which objects fly. The theory is plausible since it draws contradictory conclusions about immobile birds. Each of the influents are plausible in the sense that they have many exceptions. We will demonstrate how an agent can use the remainder of these influents to rectify one of the missing preconditions in the $Bird(x) \succ Fly(x)$ influent. For reasons external to the PI–EBL algorithm let us assume the agent needs to learn which objects are likely to fly. Suppose the first example seen by the agent is a clawed, feathered bird which can fly. This is explained using: $Clawed(x) \succ Bird(x) \succ Fly(x)$. This explanation is rejected when the agent sees a clawed, hairy bear which cannot fly. The next most a priori plausible explanation chosen is consistent with the examples seen: $Feathered(x) \succ Bird(x) \succ Fly(x)$. Even this explanation is insufficient in explaining a hunters trophy: a dead bird with feathers and claws which cannot fly. As before, the explanation is rejected and the next most a priori plausible explanation is generated. The explanation in figure 3 has an

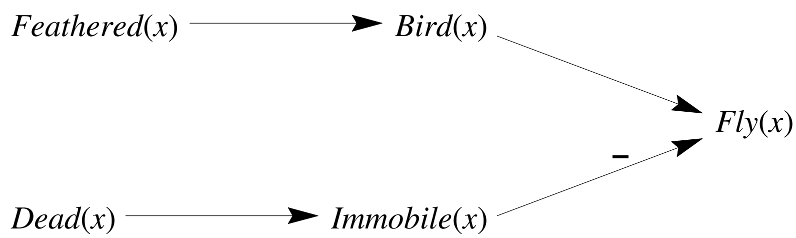

Figure 3: Plausible Explanation

influent set composed of two influences at its top level. There are two functions mapping deadness and featheredness to the ability to fly which are consistent with this plausible explanation. One predicts dead, feathered objects fly, and the second function predicts that they do not. Both are consistent with the theory but only the first is consistent with the hunter's trophy so it is accepted. There are actually sixteen functions which accept two boolean variables, but only two of the sixteen functions would cause the explanation to be deterministic, predictive, and relevant over examples consistent with the function. If the explanation is insufficient to cover some example (it is not deterministic over the examples) then a new explanation is generated, and so on, until enough examples are seen that are consistent with this explanation. At this point it is accepted as a plausible theorem along with the classification function mapping the leaves of the explanation to its consequent.

$Bird(x) \succ Fly(x)$ is missing the precondition ''not dead''. This is rectified by combining it in an explanation dealing with immobility. The missing precondition was ''added'' by building the composite explanation. If $Bird(x) \succ Fly(x)$ is later used in another explanation its simpler form may be used. This is important since it is quite likely that the dead bird exception will be irrelevant (as will the hundreds of other exceptions). As we discussed earlier the deductive version: $\ldots \wedge Dead(x) \wedge Bird(x) \succ Fly(x)$ would force the agent to reason about *all* those exceptions at *all* times. Plausible explanations allow the agent to correctly handle these cases without forcing all reasoning to be done at this level. Notice the imprecision in the plausible theory, the theory does not specify which of the two competing sub–explanations would prevail if both had their preconditions satisfied. Of course if that level of detail were known it could easily be specified. The important point is its possible to specify that featheredness relates to flying and deadness relates to flying without precisely specifying *all* possible interactions between the knowledge. The importance of this simplification increases as the theory increases in complexity because the number of possible interactions increases dramatically with complexity, and many of these interactions should not be dealt with by the agent since they will never occur in practice. Even if all necessary distinctions were known, expanding a plausible theory into a deductive theory would cause duplication. As it stands now $Dead(x) \succ Immobile(x)$ can be used as an exception to flying, walking, swimming, dancing, etc. If it were written as a deductive theory the same exception would need to be placed explicitly in the preconditions of many rules.

7. CONCLUSIONS

The advantages of knowledge intensive approaches are well known. An agent using a plausible model instead of a deductive model will retain these advantages, but like connectionist approaches avoids the unnatural constraints of consistency and precision of deductive systems. This is very important because the agent may accept and reason with new knowledge without completely understanding all possible interactions with existing knowledge. PI–EBL, because of its representation, allows very simple and imprecise theories to be made precise by learning how the plausible knowledge interacts from actual world observations (examples). In this way the agent can boot–strap itself into specialized domains by combining relevant portions of very general theories building a specialized theory which is adapted empirically.

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